

MODEL QUESTION PAPER

TED(15)-6132

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING/TECHNOLOGY

MICROCONTROLLERS

(Maximum Marks:100)

Time:3Hrs

PART A

I [Answer the following questions in one or two sentences. Each carries '2' marks]

1. Write the use of Program Counter?
2. What is RISC?
3. Name the instruction used to clear bit in I/O register?
4. Give the use of prescale operation of Timer?
5. What is DCE?

(5x2=10)

PART-B

II [Answer any five of the following questions. Each question carries '6' marks]

1. List the Features of AVR .
2. Write about conditional branch instructions
3. Write an AVR C program to get the status of bit 5 of PORTB and send it to PORT C continuously
4. List different types of interrupts.
5. Give the uses of SREG.
6. Describe the uses of DAC
7. Mention the advantages of LCD

(5x6=30)

PART-C

[Answer one full question from each unit. Each question carries '15' mark]

Module I

- III a. List and explain about different AVR family (7)
 b. Illustrate the application of stack in AVR (8)

OR

- IV a. Explain the architecture of AVR with block diagram (15)

Module II

- V. a. Explain the dual role of AVR pins (8)
 b. Explain the role of DDR register in inputting and outputting data (7)

OR

- VI a. Write a program that toggles bit 2 of PORTC (6)
 b. Write a AVR C program to read pins 1 and 0 of PORTB and issue an ASCII character
 '0' to PORTD if pin1 =0 and pin0=0
 '1' to PORTD if pin1=0 and pin0=1 (9)

Module III

- VII a. Explain the steps when an interrupt occurs to AVR (10)
 b. List different registers in Timer (5)

OR

- VIII a. Illustrate the steps to be followed to program Timer0 in CTC and Normal mode (15)

Module IV

- IX a. Explain different modes of Data transfer (7)
- b. Describe how commands are send to LCD (8)

OR

- X a. Describe the interfacing of Keyboard to AVR (10)
- b. Explain temperature sensors (5)